Multistep Methods (Adams-Bashforth & Adams-Moulton)

Group- Puppis

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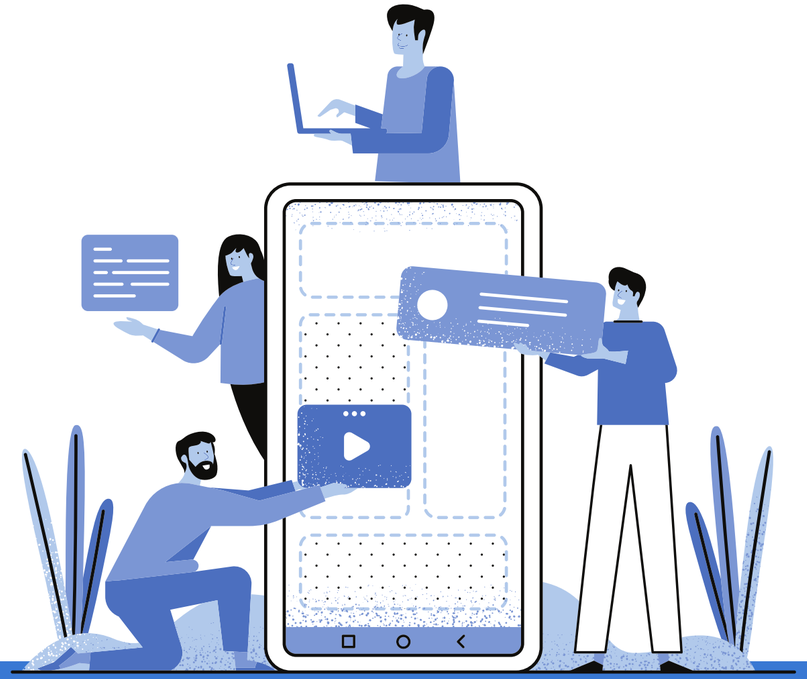

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Introduction

- Numerical methods are vital for solving Ordinary Differential Equations (ODEs) when analytical solutions are unavailable.

- Two major categories:
- Single-step methods (eg:RK)
- Multistep methods (use previous values, eg:AB,AM)

- Adams-Bashforth: Explicit multistep schemes.
- Adams-Moulton: Implicit schemes, generally more stable

Objectives of the Study

1. Implement AB2, AB3, and AM2 methods
2. Use RK4 as a reference solution
3. Compare methods using identical step sizes
4. Visualize:
   - Numerical solutions
   - Error growth
   - Stability behavior

Governing Differential Equation

General First-Order ODE: $$\frac{dy}{dt} = f(t, y), \quad y(0) = y_0$$

Test Problem: $$\frac{dy}{dt} = -2y + e^{-t}$$

- Exhibits exponential decay
- Suitable for stability analysis
- Behaves like a stiff equation

Exact Solution

The analytical (exact) solution is: $y(t) = e^{-2t} + \frac{1}{2}e^{-t}$

- Used as a reference
- Enables exact error computation
- Helps evaluate accuracy and stability

Runge-Kutta 4th Order (RK4)

- A single-step, fourth-order accurate method.

Formula: $y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

- Single-step method
- Fourth-order accurate
- Strong stability properties

Adams–Bashforth 3-Step (AB3)

AB3 Formula: $y_{n+1} = y_n + \frac{h}{12}(23f_n - 16f_{n-1} + 5f_{n-2})$

- Explicit multistep method
- Third-order accurate
- Requires previous solution values
- Highly sensitive to step size

Adams–Moulton 2-Step (AM2)

Predictor (AB2): $y_{n+1}^{(p)} = y_n + \frac{h}{2}(3f_n - f_{n-1})$

Corrector (AM2): $y_{n+1} = y_n + \frac{h}{2}[f(t_{n+1}, y_{n+1}^{(p)}) + f_n]$

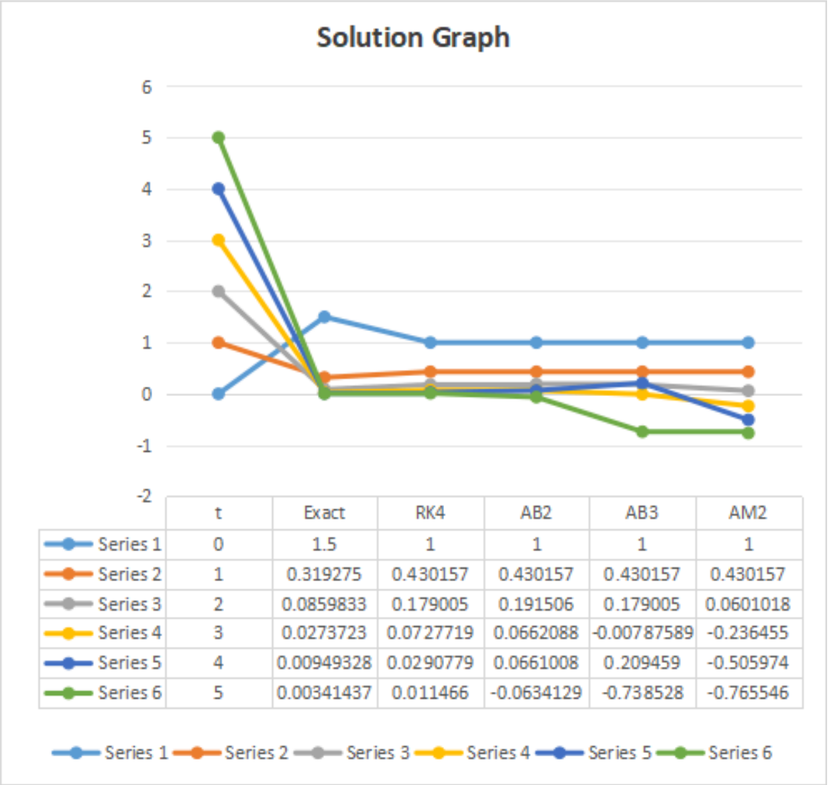
- Implicit method
- Improved stability over AB methods
- Depends strongly on predictor accuracy

Methodology

- Step Size: h = 1.0 (intentionally large to test stability).
- Time Interval: $0 \leq t \leq 5$
- Initial Value: y(0) = 1.
- Initialization: RK4 is used to generate starting values for multistep methods.
- Comparison: $\text{Error} = |y_{\text{numerical}} - y_{\text{exact}}|$

Results – Solution Behavior

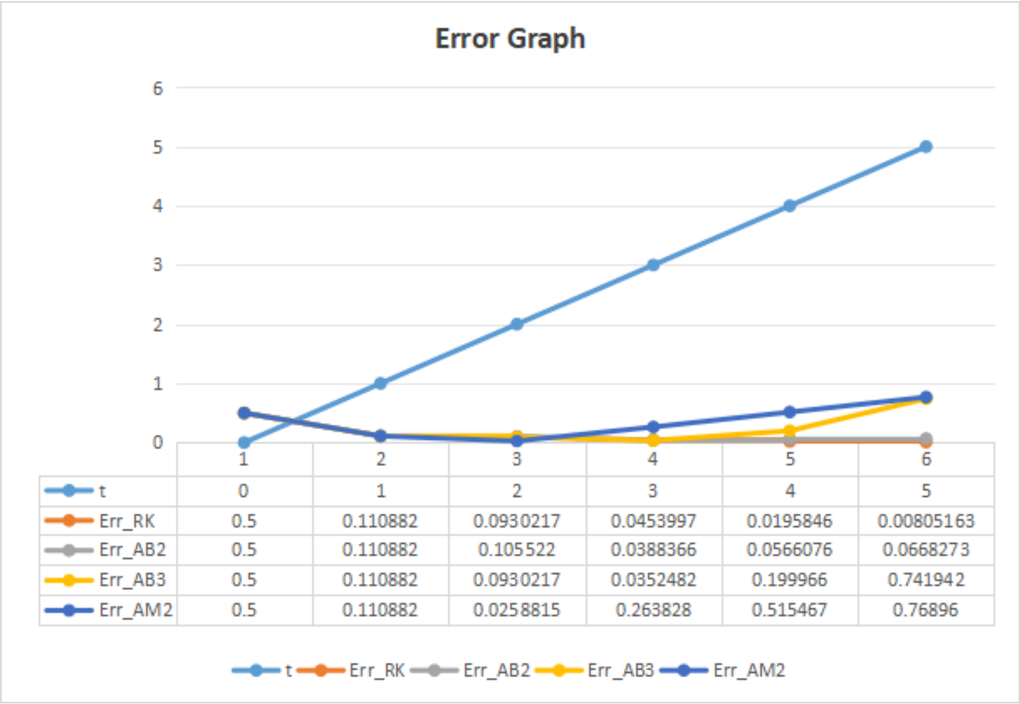
- RK4: Closely follows the exact solution throughout.
- AB2: Shows increasing deviation after t=3.
- AB3: Becomes unstable earlier, exhibiting oscillations.
- AM2: Diverges rapidly, exhibiting severe instability at large step sizes.
- Conclusion: Explicit multistep methods are highly sensitive to step size



	t	Exact	RK4	AB2	AB3	AM2
Series 1	0	1.5	1	1	1	1
Series 2	1	0.319275	0.430157	0.430157	0.430157	0.430157
Series 3	2	0.0859833	0.179005	0.191506	0.179005	0.0601018
Series 4	3	0.0273723	0.0727719	0.0662088	-0.00787589	-0.236455
Series 5	4	0.00949328	0.0290779	0.0661008	0.209459	-0.505974
Series 6	5	0.00341437	0.011466	-0.0634129	-0.738528	-0.765546

Results – Error & Stability Analysis

Method	Error
RK4	**0.00805**
AB2	0.06683
AB3	0.74194
AM2	0.76896



	1	2	3	4	5	6
t	0	1	2	3	4	5
Err_RK	0.5	0.110882	0.0930217	0.0453997	0.0195846	0.00805163
Err_AB2	0.5	0.110882	0.105522	0.0388366	0.0566076	0.0668273
Err_AB3	0.5	0.110882	0.0930217	0.0352482	0.199966	0.741942
Err_AM2	0.5	0.110882	0.0258815	0.263828	0.515467	0.76896

stability Summary:

- RK4: Most stable
- AB2: Conditionally stable
- AB3: Least stable
- AM2: Predictor-dependent instability

Conclusion

- RK4 is the most robust and reliable reference method for stability and accuracy.
- Large step sizes reveal significant stability issues in multistep methods, especially AB3.
- While computationally efficient, multistep methods require careful step-size selection to avoid divergence.

FEEL FREE TO ASK ANY QUESTION

THANK YOU